Quiz 9

1 Grading Scheme

- Question 1 is for 2 points (1 point for each part)
- Question 2 is for 1 point
- Question 3 is for 2 points (1 point for injectiviy and 1 point for surjectivity)

Total grade is out of 5.

2 Solutions

1. (a) $(3 - n)$ iterations and 4 elementary operations per iterations. Hence, the total number of operations is $4(n - 3)$.

   (b) Using the polynomial theorem, the order of the algorithm is $O(n)$.

2. $$\begin{aligned} a_n &= n^2 + n + 1 + \frac{1}{n} + \sin n \\ &\leq n^2 + n^2 + n^2 + n^2 + n^2 \\ &= 5n^2 \end{aligned}$$

   Take $n_0 = 1$ and $B = 1$. Then, $a_n \leq Bn^2$ for all $n \geq n_0$.

3. - Solution 1:
     Injectiviy: Suppose there exist $x \neq y$ such that $f(x) = f(y)$. Then, $x + 1 = y + 1$. This implies that $x = y$, a contradiction. Hence, f is injective.
     Surjectivity: Let y be any odd number. Let $x = y - 1$. Then x is even. Moreover, $f(x) = x + 1 = y$. Therefore, f is surjective.

     Therefore, f is bijective.

   - Solution 2:
     Show that there exist an inverse. Let $g : \{\text{odd integers}\} \to \{\text{even integers}\}$, given by $g(x) = x - 1$. Then $g \circ f = id$ and $f \circ g = id$. This implies that g is an inverse of f and hence, f is bijective.

3 Common error

1. Injective means $f(x) = f(y) \implies x = y$. It does not mean $x = y \implies f(x) = f(y)$. This is true for any function.

2. While proving surjectivity, pick any element y from the codomain and show that there exist an element x in the domian such that $f(x) = y$.